Local Scaling and Temporal Learning in Chart-Based Return Prediction

Abstract

We study what a chart-based convolutional neural network (chart CNN) learns from recent market histories. Chart construction combines local high–low scaling, a binary grayscale image, a fixed pixel layout, and flexible learning. We examine these components by comparing the chart CNN with logistic classifiers and one-dimensional convolutional neural networks (CNN1D) applied to the same underlying price and volume sequences under alternative scaling schemes. Local high–low scaling exposes a strong weekly return ranking to the logistic classifier before nonlinear feature extraction. CNN1D applied to the image-scaled sequence then reproduces the chart CNN's portfolio performance across input windows. Chart and sequence scores rank many of the same stocks, and multimodal fusion adds little to the stronger representation. These findings indicate that the chart primarily delivers a locally normalized temporal signal. Transparent price rules, transaction-cost adjustments, weighting schemes, and tradability screens define its economic scope. Historical performance is concentrated in equal-weighted portfolios and small, illiquid stocks. Net performance becomes negative under modest tradability screens and attenuates during 2020–2024. Chart-based predictability therefore reflects representation and temporal learning, with limited incremental performance from the chart image and limited scalable value.

Keywords: Return predictability, Machine learning, Representation learning, Technical analysis, Transaction costs

JEL classification: C45, G11, G12, G14

1. Introduction

Chart-based models raise a basic question for return prediction. When a machine learns from a market chart, what has it learned? The answer shapes how we interpret predictability from recent price and volume data. Before the model sees a chart, the input window has already been selected, locally normalized, and rendered into a visual object. Strong chart CNN performance may therefore reflect the image layout, the local scaling of the input, or the ability of a flexible model to extract nonlinear structure from price-volume information.

We study the representation question through three learned model families. The chart CNN learns from price and volume images, while CNN1D learns from scaled temporal sequences built from the same underlying data. The multimodal model combines the chart and sequence branches. CNN1D tests how much of the chart CNN signal can be recovered without the image layout, and the multimodal model tests whether the two representations contain complementary predictive information.

The sequence models are evaluated under three scaling schemes. Image scaling preserves the normalization embedded in chart construction, while cumulative-return and devolatized-return scaling provide alternative numerical representations of the same recent market history. We apply both a logistic classifier and CNN1D to these inputs. The logistic model shows how much predictive content is present in each encoding, and CNN1D shows what is added through nonlinear temporal learning.

The multimodal model provides evidence on representation complementarity. If the chart and sequence branches extract distinct predictive information, combining them should improve the stock ranking beyond either branch alone. If their information mostly overlaps, multimodal gains should be limited. We complement this comparison with score correlations, encompassing regressions, and incremental explanatory-power tests because similar portfolio performance can arise from different stock rankings.

Our empirical analysis uses daily U.S. equity data from CRSP. Model development is confined to 1993–2000, and we report out-of-sample results for the full 2001–2024 period. We divide the evaluation period into a 2001–2019 baseline sample and a 2020–2024 extension sample. We compare the learned models with momentum (MOM), one-month short-term reversal (STR), one-week short-term reversal (WSTR), and a multi-horizon trend strategy (TREND). All methods use the same universe, timing discipline, portfolio construction rules, weighting schemes, and transaction-cost treatment.

The central result is that local high–low scaling exposes a strong weekly return ranking to a linear classifier. In the representative $I20/R5$ specification, the image-scaled logistic classifier produces an equal-weighted gross Sharpe ratio of 5.22, compared with −0.41 under cumulative-return scaling. CNN1D raises the Sharpe ratio to 6.55, close to 6.52 for the chart CNN. Across weekly input windows, CNN1D matches or exceeds the chart CNN's portfolio performance.

The score-level tests indicate substantial, though incomplete, overlap. Chart and sequence rankings are most similar in the strongest weekly specifications, and the strongest individual branch retains the highest performance under multimodal fusion. The 60-day input reveals more representation-specific variation. CNN1D preserves its rankings more consistently across input windows and provides the most consistent conditional content in the joint regressions.

The learned portfolios exceed the transparent price-based benchmarks at the weekly horizon under common portfolio rules. The advantage is concentrated in equal-weighted portfolios and narrows at longer horizons and under value weighting. Full-universe weekly portfolios retain net Sharpe ratios near four after the stated transaction costs, while modest dollar-volume and market-capitalization screens produce negative net returns. The economic result is therefore strongest among small, illiquid stocks.

Performance weakens during 2020–2024 across the learned models, WSTR, and TREND. Several equal-weighted learned specifications retain positive net performance. The learned-model advantage nevertheless narrows and remains economically difficult to scale.

1.1. Related literature

Recent market history can be represented through prespecified price-based signals, ordered numerical sequences, or chart images. Lo et al. (2000) show how chart patterns can be defined and tested systematically, Jiang et al. (2023) apply a CNN to price and volume images, and Murray et al. (2024) study nonlinear predictability in numerical return histories. We hold the underlying market history fixed and vary its representation, which allows us to examine the contributions of local high–low scaling, chart layout, and temporal learning within one empirical design.

A second connection is to the interpretation of machine-learning predictions in asset pricing. Gu et al. (2020) document the predictive value of flexible nonlinear models, while Cheng et al.

(2025) show that carefully specified regressions can remain competitive. Our scaling comparisons show that local high–low scaling exposes substantial predictive content to a linear classifier before nonlinear temporal learning is added.

The third connection concerns the economic scope of short-horizon return prediction. Chen et al. (2025) show that weekly reversal is concentrated among small and illiquid stocks, Novy-Marx and Velikov (2016) document the effect of transaction costs on anomaly returns, and Kaczmarek and Pukthuanthong (2025) find that chart CNN performance weakens in newer data and under implementation constraints. We trace the learned signal across weighting schemes, transaction costs, tradability screens, and the 2020–2024 extension sample.

The paper makes one central contribution. It uses a controlled representation comparison to examine how local high–low scaling, the pixel layout, and temporal learning contribute to chart CNN predictability. Local scaling exposes a strong stock ranking to the logistic classifier, and CNN1D recovers the chart CNN's weekly portfolio performance from the same locally scaled market history. The multimodal, benchmark, transaction-cost, tradability, and sample-stability results document the overlap between representations and the economic boundaries of the signal.

Section 2 presents the empirical design. Section 3 studies how local scaling and temporal learning contribute to chart CNN predictability and whether the learned models rank the same stocks. Section 4 evaluates performance against transparent price-based benchmarks and examines the economic and sample boundaries of the signal. Section 5 concludes. The online appendix reports implementation details and the complete result grids.

2. Data, representations, and empirical design

Chart construction combines local high–low scaling, a binary grayscale image, and flexible learning. We examine these components through a controlled representation comparison. The logistic models show how scaling affects the stock ranking within a fixed linear architecture before nonlinear learning is added. The chart CNN and CNN1D then receive the same locally scaled market history, and the multimodal model tests whether their representations contain complementary predictive information. We hold timing, ranking, weighting, and transaction-cost rules fixed across the learned scores.

2.1. Sample and prediction target

To hold the data environment fixed, we construct a daily CRSP panel of ordinary common shares listed on NYSE, AMEX, or Nasdaq using share codes 10 and 11. The panel includes daily returns, open, high, low, and close prices, trading volume, lagged market capitalization, stock identifiers, and available delisting returns. At each formation date, eligibility requires positive prices and sufficient history for the relevant signal. Firms enter and leave with their CRSP histories, producing an unbalanced panel with no survivorship requirement.

We separate model development from economic evaluation. Model development uses data from 1993–2000, and all portfolio results are evaluated out of sample from 2001 to 2024. We report the full evaluation period, a 2001–2019 baseline sample, and a 2020–2024 extension sample.

Our model grid combines 5-, 20-, and 60-day input windows with 5-, 20-, and 60-day forecast horizons. We write each specification as I/R, so $I20/R5$ uses 20 trading days of market history to predict the return sign over the next five trading days. For stock i at formation date t, the directional target is defined by

$$y_{i,t}^{(R)} = \mathbb{1}\left(\prod_{j=1}^{R}(1 + r_{i,t+j}) - 1 > 0\right). \tag{1}$$

The target period begins at $t + 1$, after all input information has been observed by the close of date t.

Stock splits can create jumps in raw nominal prices even when the underlying return path is continuous. We therefore set the first close in each input window to one, reconstruct subsequent closes from daily returns, and scale open, high, and low prices consistently with the reconstructed close series. The moving-average window equals I. The chart and sequence inputs are constructed from this common price-consistent market history and trading volume.

Within each specification, the three learned models use the same underlying data. Eligible cross-sections can vary across input windows and price-based benchmarks because the signals require different amounts of historical data. Each portfolio uses the sample available for its signal under common portfolio rules. Direct comparisons of the chart CNN, CNN1D, and multimodal scores use an exact common stock-date sample. Portfolio performance and signal overlap are therefore evaluated on samples suited to their distinct questions.

2.2. *From market histories to charts*

The chart input combines local scaling with a binary grayscale representation. Local scaling removes nominal price levels and preserves the relative shape and range of each stock's recent path. Chart construction then places open, high, low, close, moving-average, and volume information in a fixed image layout processed by the chart CNN. We use the representation tests to assess the contribution of local scaling and the chart image.

For each stock and formation date, we scale the five price channels within the lookback window. Let $m_{i,t}^{(I)}$ and $M_{i,t}^{(I)}$ denote the minimum and maximum calculated jointly across all open, high, low, close, and moving-average observations in that window. Each price-channel observation $x_{i,t-j}$ is then transformed using these values,

$$\widetilde{x}_{i,t-j} = \frac{x_{i,t-j} - m_{i,t}^{(I)}}{M_{i,t}^{(I)} - m_{i,t}^{(I)}}. \tag{2}$$

Trading volume is min–max scaled separately within the same window.

After scaling, we construct the chart using daily OHLC bars, a moving-average line, and a separate lower panel for trading volume. The 5-, 20-, and 60-day images have dimensions 32×15, 64×60, and 96×180 pixels. Pixels corresponding to the OHLC bars, moving-average line, and volume bars take value one, while empty pixels take value zero. The resulting binary grayscale image in Figure 1 preserves the relative structure of recent prices and volume.

Each sequence observation contains six aligned daily series in chronological order. The channels are open, high, low, close, moving average, and trading volume. We evaluate the image scale, cumulative-return scale, and devolatized-return scale. Under image scaling, the price channels retain the continuous within-window values from Eq. (2), while volume is scaled separately. The cumulative-return scale anchors each price channel to the first close in the lookback window. The devolatized-return scale expresses price changes relative to lagged exponentially weighted volatility and volume through relative changes. We standardize each transformed

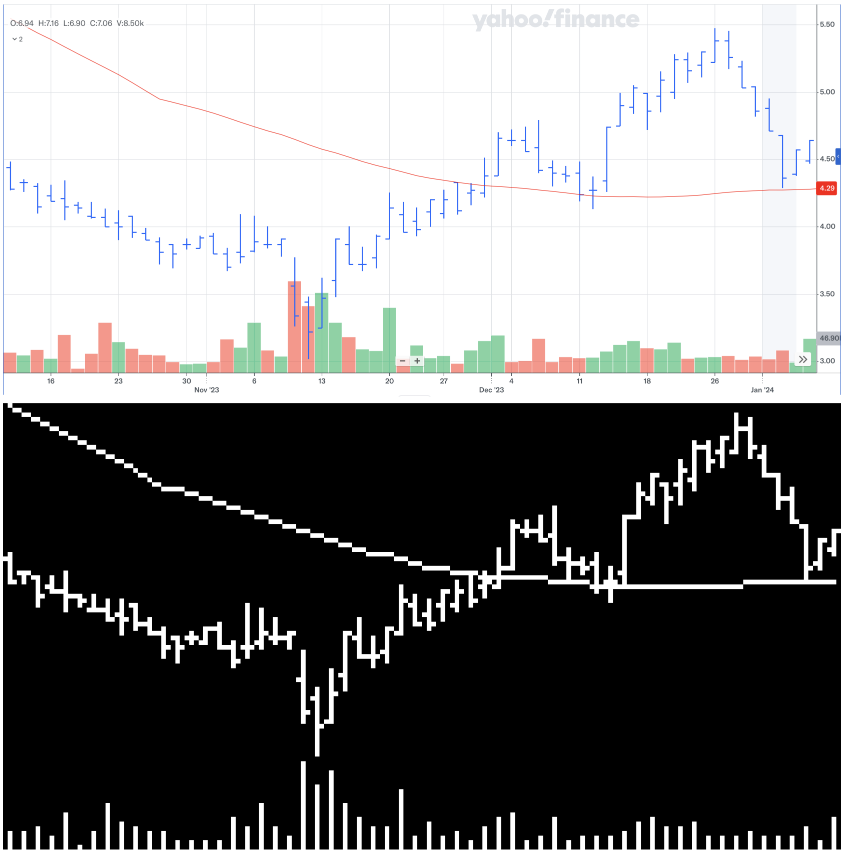

Figure 1: Example of chart construction and model input. The upper panel shows a conventional 60-day OHLC chart for Envela Corporation (ELA). The lower panel shows the corresponding binary grayscale image used by the chart CNN, with the OHLC bars and moving-average line in the price panel and trading volume in the lower panel.

channel using development-sample means and standard deviations that remain fixed throughout the evaluation period.

The chart and image-scaled sequence use the same local normalization. The sequence retains continuous open, high, low, close, moving-average, and volume observations as channels ordered by trading day. The chart converts these observations into a fixed binary pixel layout, which reduces numerical precision while preserving relative shape and location. The image-scaled sequence therefore shows how much of the chart CNN's stock ranking can be reproduced from the same locally scaled market history.

Table 1 summarizes the model inputs, scaling schemes, and model structures used in the representation comparison. The logistic classifiers use the scaled sequence inputs as linear indices and show how much predictive content each scaling scheme makes available to a linear model. CNN1D learns nonlinear temporal patterns from the same numerical channels. The chart CNN learns from the binary grayscale image, and the multimodal model combines the chart and sequence representations through paired branches.

The logistic specifications use the same directional target as CNN1D and map the flattened sequence into a linear index. Comparing them across scaling schemes measures how scaling changes the ranking within a fixed linear model. Comparing logistic and CNN1D under local scaling measures the increment associated with nonlinear interactions across adjacent observations. The chart CNN/CNN1D comparison then asks whether the binary image representation produces an economically different ranking after both models receive the same locally normalized history.

The final comparison studies complementarity. Multimodal inputs are paired by stock, formation date, input interval, and forecast horizon. A stable improvement from fusion would show that the chart and sequence branches extract differences that are useful in the portfolio sort. Similar branch and fusion performance would point toward a common economic signal. This evidence is descriptive because the fusion rule and optimization can constrain the joint model. The central interpretation therefore rests on patterns that recur across scaling schemes, input windows, model specifications, and the separate overlap tests.

The design narrows the source of performance without creating a pure treatment for layout. Chart and sequence models differ in numerical precision, dimensionality, filter geometry, depth, and capacity. We interpret similar performance after local scaling as evidence that the pixel image is unnecessary for reproducing the tested ranking. Residual differences can arise from any of the remaining representation or architecture features.

Table 1: Models and scaling schemes in the representation comparison

Model	Scaling	Input representation	Model structure
Logistic classifier	Cumulative-return scale	Continuous sequence	Linear logit
Logistic classifier	Devolatized-return scale	Continuous sequence	Linear logit
Logistic classifier	Image scale	Continuous sequence	Linear logit
CNN1D	Cumulative-return scale	Continuous sequence	Temporal convolution
CNN1D	Devolatized-return scale	Continuous sequence	Temporal convolution
CNN1D	Image scale	Continuous sequence	Temporal convolution
Chart CNN	Image scale	Binary grayscale image	Two-dimensional convolution
Multimodal model	Image scale	Chart and sequence	Paired branches with fusion

2.3. Prediction models and estimation

The chart CNN architecture follows the design used by Jiang et al. (2023). We estimate separate variants for the 5-, 20-, and 60-day chart inputs. Their depth increases with the length and resolution of the input image, with two, three, and four convolutional blocks, respectively. Each block applies a 5×3 convolution, batch normalization, Leaky ReLU activation, and max pooling. The final feature map is flattened, dropout is applied, and a fully connected layer maps the learned representation to the two return-sign classes.

We estimate separate CNN1D architectures for the 5-, 20-, and 60-day sequence inputs. Each block applies a one-dimensional convolution along the time dimension, followed by the same normalization, activation, and pooling operations as in the chart CNN. The filters span three trading days and observe all six channels at each location. The three input lengths use one, two, and three blocks, respectively. The first block has 64 output channels, and the channel count doubles in each subsequent block. After flattening, the final sequence representation passes through the same dropout and fully connected classification layers as the chart CNN.

Each multimodal observation pairs a chart image and a sequence input for the same stock, formation date, input interval, and forecast horizon. The chart CNN and CNN1D branches process their inputs separately. Let $f_{i,t}^{2D}$ and $f_{i,t}^{1D}$ denote the corresponding feature vectors. Feature concatenation forms the joint representation

$$f_{i,t}^{MM} = \left[f_{i,t}^{2D}; f_{i,t}^{1D}\right]. \tag{3}$$

The concatenated feature vector is passed to a linear classifier and the final softmax layer. This design allows each branch to learn its own representation before the classifier combines them. We also examine late fusion, where the chart CNN and CNN1D produce separate predictions that are combined through probability or logit averaging. Comparing the multimodal and single-branch models tests whether the two representations contain complementary predictive information.

The final layer of each model produces one score for the down class and one for the up class. The softmax transformation converts these scores into probabilities. The probability assigned to the up class is

$$s_{i,t} = \frac{\exp(z_{i,t}^{(1)})}{\exp(z_{i,t}^{(0)}) + \exp(z_{i,t}^{(1)})}, \tag{4}$$

We carry $s_{i,t}$ forward as a within-date ranking score for the portfolio analysis. We do not interpret it as a calibrated expected return.

All learned model families are estimated within a common prediction framework. We estimate separate models across input intervals, forecast horizons, model families, and, where applicable, sequence scaling or fusion design. The neural models use cross-entropy loss, the Adam optimizer, a learning rate of 10^{-4}, a batch size of 128, dropout probability of 0.50, Xavier initialization, and a maximum of 50 training epochs.

Training stops when validation loss does not improve for two consecutive epochs. We retain the model parameters from the epoch with the lowest validation loss as the selected checkpoint. These parameters remain fixed throughout the 2001–2024 evaluation period and produce one up-class probability for each eligible stock-date observation.

Model development uses data from 1993–2000. We randomly divide stock-date observations into 70% training and 30% validation sets using a fixed seed. Observations from the same firm or calendar period can therefore appear in both sets. Validation is used to select the checkpoint within the pre-2001 development sample. Portfolio evaluation begins in 2001, after model fitting and checkpoint selection are complete.

The main analysis uses one selected checkpoint for each learned specification. The portfolio score is therefore the single-model up-class probability from that checkpoint. For $I5/R5$ and $I20/R5$, the online appendix compares the single-model chart CNN with matched five-member forecast averages. This sensitivity check is limited to these weekly chart CNN specifications, and the weekly conclusions remain unchanged.

2.4. Portfolio construction

At each formation date, stocks are sorted according to the relevant model score or benchmark signal, and the eligible cross-section is divided into ten deciles. The high-minus-low portfolio buys the highest-score decile and shorts the lowest-score decile. We report equal-weighted and value-weighted portfolios. Equal weighting assigns the same weight to each stock within a leg, while value weighting uses lagged market capitalization normalized separately within the long and short legs. Each leg carries one dollar of notional, so the spread has zero net investment and two dollars of gross exposure.

Weekly, monthly, and quarterly portfolios correspond to the 5-, 20-, and 60-trading-day forecast horizons. At formation date t, the model score uses information available through the close of that date and ranks stocks for the return from $t+1$ through $t+R$. We use one formation date per holding interval, producing non-overlapping return series at each evaluation frequency. For a portfolio return series $R_{p,t}$ with $K \in \{52, 12, 4\}$ observations per year, the reported Sharpe ratio is

$$\mathrm{SR}_p = \sqrt{K}\,\frac{\overline{R}_p}{\mathrm{sd}(R_{p,t})}. \tag{5}$$

No risk-free rate is subtracted from the zero-net-investment spread.

Benchmark signals can require longer histories than the learned models, which produces different eligible cross-sections. All methods use common ranking, horizon, weighting, and transaction-cost rules within the same CRSP universe. Comparisons among the chart CNN, CNN1D, and multimodal model use the exact common stock-date sample described in Section 2.1.

We report the complete grid across input intervals and forecast horizons. Later figures and economic tests use $I20/R5$ as the representative weekly specification. It is central to the short-horizon grid, performs strongly across all three learned model families, and allows the models to be compared under the same horizon structure. We selected this specification after examining the grid, so the focused results are descriptive.

2.5. *Price-based benchmarks*

We compare the learned models with four transparent price-based benchmark strategies. These are momentum (MOM), one-month short-term reversal (STR), one-week short-term reversal (WSTR), and the multi-horizon trend strategy (TREND). MOM captures continuation, STR and WSTR capture reversal at different horizons, and TREND captures broader multi-horizon trend information. All four strategies are evaluated under the common portfolio rules described above.

MOM is the medium-horizon continuation benchmark. Following Jegadeesh and Titman (1993), we implement a daily analogue of a 12–1 momentum strategy. The signal is the cumulative return over the previous 252 trading days, excluding the most recent 21 trading days,

$$z_{i,t}^{\mathrm{MOM}} = \prod_{j=22}^{273} (1 + r_{i,t-j}) - 1. \tag{6}$$

The exclusion of the most recent 21 trading days separates medium-horizon continuation from the short-term reversal patterns captured by STR and WSTR.

One-month short-term reversal (STR) follows Jegadeesh (1990) and uses the negative of the previous 21-trading-day return. One-week short-term reversal (WSTR) follows Lehmann (1990) and uses the negative of the previous 5-trading-day return.

$$z_{i,t}^{\mathrm{STR}} = -\left(\frac{P_{i,t}}{P_{i,t-21}} - 1\right), \qquad z_{i,t}^{\mathrm{WSTR}} = -\left(\frac{P_{i,t}}{P_{i,t-5}} - 1\right). \tag{7}$$

High scores identify recent losers. The price series used for both signals is split-consistent with the return history. WSTR provides the closest transparent benchmark to the learned weekly forecast horizon.

TREND follows the multi-horizon trend strategy of Han et al. (2016) and combines moving-average information across several horizons. For each lag L in the set

$$\mathcal{L} = \{3, 5, 10, 20, 50, 100, 200, 400, 600, 800, 1000\}$$

we define $x_{i,t,L} = \mathrm{MA}_{i,t,L}/P_{i,t}$, which compares the moving average with the current price. At each formation date, a cross-sectional regression maps these signals into next-period returns. The forecasting coefficients are the trailing mean of the previous 12 fully realized coefficient vectors. The date-t TREND score is

$$z_{i,t}^{\mathrm{TREND}} = \sum_{L\in\mathcal{L}} \overline{\beta}_{L,t} x_{i,t,L}. \tag{8}$$

TREND is a supervised linear benchmark with explicit features and lagged coefficient estimation.

2.6. *Cost adjustment and implementation tests*

We report both gross and net portfolio performance. At each rebalancing date, turnover for each leg is measured from the change between its previous and target portfolio weights. For $\ell \in \{\text{long}, \text{short}\}$, turnover is

$$\mathrm{Turnover}_t^{\ell} = \frac{1}{2}\sum_i \left| w_{i,t}^{\ell,\mathrm{target}} - w_{i,t}^{\ell,\mathrm{prev}} \right|. \tag{9}$$

Total turnover is the sum of turnover in the long and short legs and is reported as a monthly-equivalent average. Net performance applies a one-way transaction cost to the full absolute traded notional in both portfolio legs. This traded notional equals twice the conventional half-turnover measure used for reported turnover. Stocks above the contemporaneous NYSE 80th-percentile market-cap breakpoint are charged 10 basis points, while the remaining stocks are charged 20 basis points. Net return is

$$R_{p,t}^{\mathrm{net}} = R_{p,t}^{\mathrm{gross}} - \sum_{i,\ell} c_{i,t} \left| \Delta w_{i,t}^{\ell} \right|. \tag{10}$$

The common cost schedule provides a transparent implementation test. It does not account for stock-specific spreads, price impact, borrowing costs, short-sale constraints, or capacity limits.

We re-form the representative $I20/R5$ weekly portfolios under five tradability screens. The screens require a closing price of at least \$5, dollar volume of at least \$1 million or \$5 million, market capitalization above the NYSE 20th-percentile breakpoint, or the combined price, \$5 million dollar-volume, and market-capitalization conditions. Dollar volume equals closing price times trading volume at the formation date. The tests use the same saved model scores as the main analysis and leave the model parameters fixed.

We obtain daily market, size, and value factors and the daily momentum factor from the Kenneth R. French Data Library. Each factor is compounded over the weekly holding interval. We then estimate Carhart four-factor regressions for weekly net high-minus-low returns (Carhart 1997). The intercepts are annualized, and inference uses Newey–West standard errors with four lags.

2.7. *Signal overlap and conditional content*

The portfolio tests show whether the models produce economically useful rankings. We also examine whether the learned-model rankings contain distinct information. For each input interval and forecast horizon, we form an exact common stock-date panel on which all three learned-model scores and future returns are observed. We compute date-level Spearman rank correlations (Spearman 1904) and average them through time. These rank correlations compare the full cross-sectional orderings produced by the models.

The second diagnostic asks which learned-model scores retain predictive content when they are observed jointly. At each formation date, we estimate a cross-sectional regression of future returns on all three scores using the exact common stock-date panel (Fama and MacBeth 1973),

$$R_{i,t+1:t+R} = \alpha_t + \beta_t^{2D} s_{i,t}^{2D} + \beta_t^{1D} s_{i,t}^{1D} + \beta_t^{MM} s_{i,t}^{MM} + \varepsilon_{i,t+R}. \tag{11}$$

The Fama–MacBeth coefficients are the time-series averages of the date-level slopes. The conventional test statistics use the time-series standard errors of these slopes without an autocorrelation correction. We therefore treat the significance levels as exploratory, particularly at longer horizons.

A positive coefficient in the joint regression indicates that the corresponding model score contains predictive variation that is not fully absorbed by the other learned-model scores. The regression provides an incremental-content diagnostic within the exact common stock-date panel. Because the scores are up-class probabilities from separately trained models, their numerical scales can differ, and correlation among the scores can affect the coefficient magnitudes. We therefore focus on coefficient signs and patterns that recur across specifications. Rank correlations and portfolio sorts provide the direct evidence on ranking similarity and economic performance.

The model grid contains related comparisons across input intervals, forecast horizons, scaling schemes, model architectures, fusion designs, weighting schemes, and sample periods. We selected the representative $I20/R5$ specification after examining the grid and do not apply a formal multiple-testing correction. We therefore base the interpretation on patterns that recur across the grid and on the separate benchmark, transaction-cost, tradability, and extension-sample evidence. The encompassing regressions provide exploratory evidence on conditional predictive content.

3. What the chart CNN learns

This section examines how local scaling and temporal learning contribute to the learned return ranking. We first compare logistic classifiers across the three sequence scaling schemes to measure how much predictive content is exposed by normalization. We then compare image-scaled CNN1D with the chart CNN under common input intervals and forecast horizons. Finally, the multimodal model tests whether the chart and sequence representations contain complementary predictive information.

3.1. Local scaling exposes the weekly signal

Table 2 reports annualized gross Sharpe ratios for weekly high-minus-low portfolios over 2001–2024. Panel A compares the logistic classifier under the cumulative-return scale, devolatized-return scale, and image scale, showing how scaling affects the ranking before temporal learning is introduced. Panel B compares image-scaled CNN1D, the chart CNN, and the multimodal model across the three input intervals and both weighting schemes. Complete monthly and quarterly results appear in the online appendix.

Panel A shows that local scaling exposes the first component of weekly predictability. In the representative $I20/R5$ specification, the logistic Sharpe ratio increases from −0.41 under the cumulative-return scale to 5.22 under the image scale, with the devolatized-return scale between them. The same ordering holds across the other input intervals. Local high–low normalization

Table 2: Local scaling and model architecture in weekly return prediction

	5-day input		20-day input		60-day input	
Model	EW	VW	EW	VW	EW	VW
Panel A. Encoding with a logistic classifier						
Logistic, cumulative scale	0.18	−0.04	−0.41	−0.51	−0.26	−0.53
Logistic, devolatized scale	2.80	0.79	2.73	0.48	2.70	0.66
Logistic, image scale	4.88	1.25	5.22	1.09	5.44	1.12
Panel B. Flexible models with image scale						
CNN1D	6.24	1.16	6.55	1.12	5.89	2.06
Chart CNN	5.88	1.39	6.52	1.44	3.72	1.18
Multimodal	6.01	1.16	6.48	1.52	5.53	1.75

Note. The table reports annualized gross Sharpe ratios of non-overlapping 5-day high-minus-low decile portfolios from 2001 through 2024. EW and VW denote equal- and value-weighted legs. Image scale applies the chart's within-window min–max transformation to continuous sequence channels. Chart CNN receives the corresponding binary image. Multimodal is the feature-concatenation model. Each row uses that model's available score cross-section; the rows are not restricted to the exact three-model stock-date intersection used in the overlap tests. The online appendix reports the full scaling and return-horizon grids.

therefore exposes a strong return ranking before temporal learning is introduced. The next comparison measures the additional contribution of temporal convolution.

Image scaling jointly normalizes open, high, low, close, and moving-average values within each lookback window, while volume is scaled separately. The transformation removes nominal price levels and preserves the relative positions of prices, ranges, and the moving average within recent market history. A given price movement occupies a larger share of the normalized range in a narrow-range window and a smaller share in a wide-range window. The image-scaled sequence retains this information as continuous channels without the pixel layout. Its logistic performance shows that chart-style normalization already provides an economically meaningful representation of recent price history.

3.2. What temporal learning adds

Panel B shows that temporal convolution applied to the image-scaled sequence reproduces the chart CNN's weekly portfolio performance. In the representative $I20/R5$ specification, the Sharpe ratio increases from 5.22 for the image-scaled logistic classifier to 6.55 for CNN1D, which is nearly identical to 6.52 for the chart CNN. CNN1D remains competitive across the other input intervals. Together, Panels A and B point to two components of performance. Local scaling exposes the baseline ranking, and temporal convolution strengthens it. The normalized numerical sequence is sufficient to reproduce the central chart CNN result.

The longer-horizon grid shows an important boundary. Image scaling remains the strongest logistic representation, but temporal convolution adds little once the forecast horizon extends beyond one week. In $I20/R20$, the image-scaled logistic classifier reaches a Sharpe ratio of 2.37, compared with 2.15 for CNN1D, and the difference becomes more pronounced as the quarterly signal weakens. Local scaling therefore remains informative across horizons, while the additional contribution of temporal learning is concentrated at the weekly horizon.

Value weighting provides a second boundary on the weekly result. In $I20/R5$, Sharpe ratios near 6.5 for the three equal-weighted learned portfolios fall to between 1.12 and 1.52 under value weighting. The models remain close to one another, preserving the representation result, and the smaller economic magnitude indicates that much of the weekly spread comes from stocks receiving little weight in value-weighted portfolios. The later tradability screens examine this concentration directly.

The representative $I20/R5$ decile profiles show that the weekly performance reflects a broad and similar ranking across the three learned models. Equal-weighted annualized returns rise from approximately −32% to −35% in the lowest-score decile to 53% to 55% in the highest, with generally increasing returns through the middle deciles. The especially steep increase from decile 9 to decile 10 shows additional concentration in the highest-ranked tail. Value-weighted profiles are flatter and receive little contribution from the lowest-score portfolio, reinforcing the smaller-stock boundary identified above.

Table 2 shows the portfolio-level result. Image-scaled CNN1D reproduces the chart CNN's central weekly performance using the continuous sequence representation. The models retain meaningful differences in input precision, filter geometry, and depth, which can affect their individual stock scores. The remaining question is whether their stock rankings are also similar. The score-overlap tests address this directly.

3.3. Do the models rank the same stocks?

The similar portfolio performance of the learned models raises the question of whether they rank the same stocks. Table 3 reports average date-level Spearman rank correlations between model scores on the common stock-date panel. These correlations measure how similarly the models order stocks within each formation date and connect the portfolio results to overlap in the underlying ranking signals.

Table 3: Rank overlap across learned models and input windows

Specification	Chart/CNN1D	Chart/Multimodal	CNN1D/Multimodal
Panel A. Across-model rank overlap			
$I5/R5$	0.73	0.74	0.84
$I20/R5$	0.67	0.72	0.80
$I60/R5$	0.44	0.65	0.69
$I5/R20$	0.50	0.67	0.71
$I20/R20$	0.53	0.65	0.72
$I60/R20$	0.44	0.55	0.63
$I5/R60$	0.45	0.45	0.68
$I20/R60$	0.44	0.45	0.64
$I60/R60$	0.38	0.46	0.63
Panel B. Within-model rank stability for the 5-day return horizon			
Model	$I5/I20$	$I5/I60$	$I20/I60$
Chart CNN	0.60	0.22	0.34
CNN1D	0.64	0.54	0.71
Multimodal	0.56	0.44	0.59

Note. Panel A reports time-series averages of formation-date Spearman correlations on the common stock-date panel for which all three learned scores and the future return are observed. Panel B compares scores from two input windows within the same model family. All statistics use the 2001–2024 evaluation window.

Rank overlap is strongest in the weekly specifications where portfolio performance is strongest. For $I5/R5$ and $I20/R5$, pairwise Spearman correlations range from 0.67 to 0.84. The similar weekly portfolio results are therefore accompanied by substantial overlap in the underlying stock rankings.

The 60-day input provides the clearest boundary to the common weekly pattern. In $I60/R5$, the chart CNN/CNN1D rank correlation falls to 0.44 as their Sharpe ratios separate to 3.72 and 5.89. Panel B further shows that CNN1D preserves its weekly ranking more consistently across input windows than the chart CNN. The longer lookback window therefore reveals a representation difference that is less visible with the 5- and 20-day inputs.

The rank correlations describe how similarly the models order stocks. Table 4 examines conditional predictive content by regressing future returns jointly on the three model scores. The reported Fama–MacBeth t-statistics show whether each score remains positively related to future returns after conditioning on the other learned-model scores.

Table 4: Joint encompassing regressions for learned-model scores

Specification	Chart CNN	CNN1D	Multimodal	Average R^2 (%)
$I5/R5$	6.95	12.56	8.39	0.70
$I20/R5$	9.51	11.80	11.91	0.80
$I60/R5$	−1.12	18.19	9.95	0.95
$I5/R20$	−0.26	4.39	−0.73	0.52
$I20/R20$	3.93	5.76	3.36	0.65
$I60/R20$	−0.85	4.66	0.28	1.06
$I5/R60$	−0.74	2.16	2.56	0.50
$I20/R60$	−0.50	2.83	1.37	0.75
$I60/R60$	−0.46	3.97	0.71	1.13

Note. The first three columns report the conventional Fama–MacBeth t-statistic for each score in a joint date-level cross-sectional regression over 2001–2024. The statistic uses the time-series standard error $\mathrm{sd}(\beta_t)/\sqrt{T}$ without an autocorrelation correction. It is therefore exploratory, especially at longer horizons and in view of the model search. The last column is the time-series average of the date-level joint-regression R^2.

The regressions sharpen the overlap result. In $I5/R5$ and $I20/R5$, all three scores retain positive conditional associations, showing that substantial rank overlap coexists with predictive variation in each score. In $I60/R5$, the chart statistic becomes weak while CNN1D remains large, consistent with the earlier separation in rankings and portfolio performance. CNN1D is positive throughout the full horizon grid and provides the most consistent conditional evidence.

3.4. *Can fusion improve the common signal?*

Fusion tests whether differences between the chart and sequence scores improve a single prediction. In the central $I20/R5$ specification, the multimodal model reaches a Sharpe ratio of 6.48, compared with 6.52 for the chart CNN and 6.55 for CNN1D. The best single branch also remains strongest in the other weekly specifications. The fusion result therefore points to limited complementarity between the two representations.

The same pattern appears when the branches are combined after separate training. Late fusion averages the chart CNN and CNN1D probabilities or logits and produces weekly performance close to the individual branches. The agreement between feature concatenation and late fusion shows that limited gains persist across fusion designs. Detailed comparisons appear in the online appendix.

Taken together, the multimodal results point to limited complementarity. Fusion preserves the strong weekly signal across feature-level and prediction-level designs, while the best individual branch retains the highest performance. Combined with the score correlations, this pattern indicates that the chart and sequence representations recover largely overlapping predictive information from the same recent price history. The joint regressions show that some conditional variation remains in the individual scores.

4. Economic performance and its limits

The representation results show that several learned models recover a strong short-horizon ranking from locally scaled recent price history. We next assess the economic scope of this ranking through transparent benchmark and implementation tests. The analysis compares the learned portfolios with price-based benchmarks, examines performance after transaction costs, locates where the returns reside in the cross section, and evaluates stability during the 2020–2024 extension sample.

4.1. Comparison with transparent price rules

Table 5 places the representative $I20/R5$ learned portfolios against MOM, STR, WSTR, and TREND over 2001–2024. Each method retains its own signal definition and lookback, and all signals enter the same weekly non-overlapping decile construction, equal-weighting rule, and transaction-cost adjustment. This common portfolio framework makes their economic performance comparable after signal construction.

Table 5: Weekly equal-weighted learned models and price-based benchmarks

Signal	Gross return	Gross SR	Net return	Net SR	Turnover
Chart CNN $I20/R5$	84.9%	6.52	51.3%	3.94	6.64
CNN1D $I20/R5$	89.1%	6.55	56.5%	4.16	6.46
Multimodal $I20/R5$	87.5%	6.48	54.5%	4.04	6.56
MOM	−2.3%	−0.08	−7.1%	−0.25	0.95
STR	44.3%	1.83	27.3%	1.13	3.32
WSTR	59.3%	2.86	24.9%	1.20	6.71
TREND	52.7%	2.49	27.5%	1.31	4.95

Note. Returns and Sharpe ratios are annualized. Turnover is monthly-equivalent conventional half-turnover summed across the two legs. The transaction-cost deduction uses full absolute traded notional. Learned and benchmark strategies draw from the same CRSP source universe, though method-specific lookback requirements can change eligible stock counts.

The benchmark comparison sets a meaningful gross-performance hurdle. WSTR is the strongest benchmark with a Sharpe ratio of 2.86, and both TREND and STR also produce positive weekly spreads. The learned models reach Sharpe ratios near 6.5 under the same portfolio framework. Combined with the representation results, this shows that models built on local scaling and temporal learning deliver substantially stronger gross portfolio performance than the individual price rules.

Trading intensity is similar for the learned portfolios and WSTR, with reported turnover between 6.46 and 6.71. The common transaction-cost schedule therefore reduces their annualized returns by approximately 33 to 35 percentage points. The larger learned gross spreads leave net Sharpe ratios near four, while TREND becomes the strongest net benchmark at 1.31. The learned net advantage reflects stronger gross rankings despite comparable turnover.

Value weighting provides a stricter implementation test. The $I20/R5$ learned portfolios retain positive gross Sharpe ratios, yet all three net annualized returns become negative, ranging from

−7.3% to −1.5%. The strong weekly net performance therefore depends on equal weighting. The tradability screens next show which stocks drive this dependence.

The benchmark comparison shows that the learned portfolios produce stronger gross and net performance than the transparent price-based rules under a common portfolio framework. Each method retains its own lookback, so eligible samples can differ across rows. A direct signal-overlap test would require exact benchmark-by-model stock-date panels and is outside this comparison. The common transaction-cost schedule captures direct trading costs, while scalability also depends on market impact, borrowing costs, short-sale constraints, and capacity. The following tradability screens provide a partial test of these implementation limits.

Across the full benchmark grid, the dominant pattern is concentration. Local scaling and temporal learning deliver their clearest advantage in weekly equal-weighted portfolios. Learned and transparent strategies converge as the forecast horizon lengthens and portfolio weights shift toward larger firms. The economic advantage is therefore concentrated in frequent sorting and portfolios that give smaller stocks more weight.

4.2. Where does the portfolio performance reside?

Table 6 examines whether the strong equal-weighted weekly results depend on stocks that are difficult to trade at scale. We re-form the representative $I20/R5$ portfolios after applying price, dollar-volume, and market-capitalization screens while keeping the model scores fixed. Applying each filter before sorting preserves the ranking rule within the eligible cross section.

Table 6: Tradability and factor diagnostics for representative weekly portfolios

Model	Screen	Net return	Net SR	Carhart α	$t(\alpha)$	Names	Turnover
Chart CNN	Full universe	51.3%	3.94	49.3%	11.81	420	6.65
	Price ≥ \$5	10.1%	1.08	8.1%	3.67	321	6.66
	Dollar volume ≥ \$1m	−10.0%	−0.85	−11.0%	−3.87	241	7.01
	Dollar volume ≥ \$5m	−19.5%	−1.57	−20.6%	−7.27	164	7.07
	No NYSE microcaps	−17.0%	−1.60	−18.1%	−7.38	193	6.89
	Combined	−22.9%	−2.10	−24.0%	−10.21	149	7.02
CNN1D	Full universe	56.5%	4.16	55.2%	12.56	419	6.46
	Price ≥ \$5	13.7%	1.48	12.4%	5.46	319	6.50
	Dollar volume ≥ \$1m	−5.7%	−0.46	−6.2%	−2.21	240	6.68
	Dollar volume ≥ \$5m	−15.0%	−1.17	−15.7%	−5.61	164	6.75
	No NYSE microcaps	−14.8%	−1.37	−15.6%	−6.34	192	6.57
	Combined	−19.0%	−1.68	−19.8%	−8.11	149	6.69
Multimodal	Full universe	54.5%	4.04	53.2%	12.10	419	6.56
	Price ≥ \$5	13.5%	1.52	12.4%	5.46	319	6.63
	Dollar volume ≥ \$1m	−5.2%	−0.42	−5.5%	−1.91	240	6.83
	Dollar volume ≥ \$5m	−14.1%	−1.11	−14.7%	−5.12	164	6.92
	No NYSE microcaps	−14.0%	−1.33	−14.6%	−5.85	192	6.75
	Combined	−19.3%	−1.76	−19.9%	−8.32	149	6.87

Note. The table reports equal-weighted $I20/R5$ portfolios from 2001 through 2024. Names is the average number of stocks in each leg. Combined requires price of at least \$5, dollar volume of at least \$5 million, and market capitalization above the NYSE 20th-percentile breakpoint. Carhart intercepts are annualized and use Newey–West standard errors with four lags.

The full-universe rows provide a common baseline for the tradability tests. All three learned models retain strong net performance under the common transaction-cost schedule, and turnover remains similar across them. With the model scores fixed, changes across the filtered rows show how restricting the eligible cross section affects economic performance.

Table 6 shows that liquidity defines the main economic boundary. Requiring a price of at least \$5 preserves positive, though much weaker, net performance, while a \$1 million dollar-volume screen makes all three portfolios negative. Excluding microcaps or combining the screens leads to the same conclusion, and turnover remains high throughout. The strong unrestricted performance is therefore concentrated among stocks that are difficult to trade at scale.

Carhart adjustment leaves the full-universe result largely unchanged, indicating that conventional market, size, value, and momentum exposures explain little of the unrestricted spread. Performance turns negative under the dollar-volume and market-capitalization screens. Together, these findings show that conventional factor exposures explain little of the signal, while its economic performance is concentrated among stocks that are difficult to trade at scale.

The tradability result places the learned signal in the same broad part of the cross section as weekly reversal. Prior evidence associates weekly reversal with temporary price pressure and finds its largest profits among small and illiquid stocks (Chen et al. 2025). Our learned portfolios are strongest in that same market segment. This common economic location motivates a connection to short-horizon reversal, while direct signal overlap remains untested.

4.3. Does the signal persist after 2019?

The extension sample tests whether the historical ranking persists in recent data with model parameters, benchmark definitions, and portfolio rules held fixed. Table 7 compares the representative weekly learned models and price-based benchmarks between the 2001–2019 baseline sample and the 2020–2024 extension sample. The shorter extension period provides a recent stability test with greater sampling uncertainty.

Table 7: Weekly equal-weighted performance in the baseline and extension periods

	2001–2019				2020–2024			
Signal	Gross ret.	Gross SR	Net ret.	Net SR	Gross ret.	Gross SR	Net ret.	Net SR
Chart CNN $I20/R5$	96.3%	7.64	60.0%	4.78	50.7%	3.87	18.5%	1.35
CNN1D $I20/R5$	102.2%	8.09	66.5%	5.28	49.9%	3.34	18.6%	1.18
Multimodal $I20/R5$	97.6%	7.76	64.4%	5.12	49.5%	3.19	17.2%	1.11
MOM	−5.1%	−0.19	−10.0%	−0.36	8.7%	0.26	4.1%	0.12
STR	48.7%	2.04	31.6%	1.33	27.7%	1.10	11.0%	0.44
WSTR	66.2%	3.33	31.7%	1.59	32.9%	1.42	−0.8%	−0.03
TREND	64.4%	3.09	38.3%	1.85	8.1%	0.38	−13.3%	−0.62

Note. Returns and Sharpe ratios are annualized. The learned rows use the representative 20-day input. Benchmark rows use the signal definitions in Section 2.5. Portfolio and cost conventions are fixed across periods.

The extension sample moves all three learned models into a common lower-performance range. Net Sharpe ratios fall from around five in the baseline sample to approximately 1.2 after 2019, while net annualized returns remain positive near 18%. The parallel decline is consistent with the shared-signal interpretation and shows that its historical economic magnitude is substantially weaker in recent data.

The recent decline extends beyond the learned models. WSTR and TREND move from net Sharpe ratios above 1.5 in the baseline sample to zero or negative values after 2019, and STR also weakens. MOM remains economically small and moves in the opposite direction. The broader deterioration points to weaker short-horizon price-history predictability across methods. The short extension leaves market change, sampling variation, and specification search as competing explanations.

The positive equal-weighted extension returns have narrow implementation scope. The same strategies produce negative net returns under value weighting, and the full-sample screens locate the unrestricted performance among stocks that are difficult to trade at scale. Taken together, the evidence shows that a detectable ranking persists in recent data, with limited scalable economic value.

The extension result aligns with the weaker recent chart CNN performance documented by Kaczmarek and Pukthuanthong (2025) and with broader evidence that return predictors decay out of sample and after publication (McLean and Pontiff 2016). Our evidence broadens this boundary because the chart CNN, CNN1D, multimodal model, and short-horizon benchmarks weaken together. The attenuation therefore spans representations and transparent price-based strategies.

The period averages conceal substantial annual variation. The annual chart CNN results in the online appendix show 2020 as the clear low point, followed by partial recovery during 2021–2024. The recent-data boundary is therefore substantial and uneven across the extension years.

4.4. Economic interpretation

The mechanism result is simple. Local high–low scaling exposes a strong weekly ranking, and temporal convolution strengthens it. The image-scaled sequence expresses prices, ranges, and the moving average relative to each stock's recent high–low range, with volume scaled separately. CNN1D reproduces the chart CNN's portfolio performance from this continuous representation, while score overlap and limited fusion gains indicate that the models recover largely shared information.

The ranking is strongest over the next week and among small, illiquid stocks. This places the learned signal in the same broad economic segment as weekly reversal (Chen et al. 2025). The connection concerns economic location, while direct score overlap remains untested.

The factor and tradability tests give different readings of the same portfolios. Strong full-universe Carhart alphas show that conventional factors explain little of the ranking. Negative returns under modest dollar-volume screens show that this statistical predictability has limited scalable portfolio value. Market impact, borrowing constraints, and capacity are also likely to be most severe among the same stocks, so the liquidity evidence defines the practical economic boundary.

We place the greatest weight on patterns that recur across model families, input windows, and economic tests. Local scaling and the additional weekly contribution of temporal learning meet this standard. Exact differences between the chart CNN and CNN1D are more sensitive to the use of one selected checkpoint per specification, differences in eligible samples, and selection from a larger model grid. Post-2019 magnitudes also come from a short extension sample. These considerations support a conservative interpretation of individual cells.

5. Conclusion

Chart CNN predictability reflects both the construction of the chart input and the learning applied to it. Each chart rescales recent prices within the stock-window and combines price, moving-average, and volume information in a binary grayscale image. We compare local scaling, the chart representation, and temporal learning to trace how each contributes to the return ranking.

Local scaling exposes the first component of the chart CNN result. The logistic classifier becomes strongly predictive when the continuous sequence receives the image scale, showing that the transformation exposes a return ranking before nonlinear temporal learning. CNN1D then strengthens the weekly ranking and reproduces the chart CNN's performance from the same image-scaled sequence. The comparison points to local normalization as the chart's main contribution, with temporal learning adding to the ranking.

The image scale makes price positions, ranges, and the moving average comparable within each stock's recent history, with volume scaled separately. This transformation removes the nominal price level and adapts the input to local price variation before learning begins. Temporal convolution then learns how these normalized relationships evolve across days. The horizon results show that scaling remains informative across forecast horizons while the additional CNN1D gain is concentrated at the weekly horizon.

The score-level tests provide a more nuanced picture. Chart and sequence rankings overlap substantially in the strongest weekly specifications, while the joint regressions still show conditional variation in the individual scores. The multimodal model performs strongly, but the best individual branch retains the highest performance under both feature concatenation and late fusion. Together, these results indicate substantial shared information and limited complementarity at the portfolio level.

The learned advantage is economically concentrated. It exceeds strong reversal and trend benchmarks at the weekly horizon under common portfolio rules, then narrows as the forecast horizon lengthens and portfolio weights shift toward larger firms. Factor adjustment leaves the unrestricted spread largely intact, while modest dollar-volume and market-capitalization screens turn net performance negative. The historical ranking is therefore strong, with limited scalable value in more tradable universes.

The extension sample adds a temporal boundary to the result. Learned models and short-horizon benchmarks weaken together after 2019, and the annual chart CNN evidence shows a sharp low point in 2020 followed by partial recovery. The attenuation spans representations and transparent strategies and remains uneven across years. The short extension leaves market change, sampling variation, and specification search as competing explanations.

Our results show why representation design is central to the interpretation of machine-learning evidence. Predictive performance reflects both the information in recent price history and the transformation applied before learning. By varying scaling, representation, and model architecture under a common prediction and portfolio framework, we trace how the return ranking changes across model designs. For the chart CNN, local normalization exposes the ranking and temporal learning strengthens it, while the chart representation provides limited incremental portfolio performance.

Further research can test the stability and economic scope of these results. Direct score comparisons with reversal and trend strategies could show how much of the locally normalized ranking they capture. Repeated model estimation and a longer extension sample could assess whether the representation comparison persists across estimation runs and market periods. More detailed estimates of market impact, borrowing costs, and capacity could determine whether the signal retains value under stricter implementation assumptions.

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